

Design and Optimization of a Distributed Low-Visibility Mesh Network in Contested Environments

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Abstract—Modern field teams operating in contested electromagnetic environments require low-profile, energy-efficient computing and communication nodes to maintain situational awareness and support local network operations. Traditional high-power radios and centralized command links are effective but often compromise stealth, consume excessive energy, are vulnerable to physical tampering, or become degraded under jamming. This practitioner paper presents the design and prototyping of a modular, low-visibility network of edge computing nodes intended for cyber sensing and local network interaction in bandwidth-denied or spectrum-constrained environments. The system integrates a single-board computer platform with a multi-layer communication architecture that includes a cellular uplink and long-range low-power radio mesh. To save power and reduce hardware costs, we separate the communication module from the main computing load, using simple microcontroller relays alongside more capable edge processors. Secure peer-to-peer communication is enabled through an open-source cryptographic routing protocol. In parallel, we present a modeling approach for node placement that frames coverage, connectivity, energy efficiency, and physical/radio-frequency exposure as a constrained set-cover and graph-connectivity optimization problem. Experimentation will include evaluating performance across open, wooded, and urban terrain and validating assumptions about communication reliability and stealth. The research demonstrates a scalable approach for resilient, low-profile cyber and sensing networks for field deployment.

Keywords—*edge computing, mesh networking, contested communications, cyber-physical systems, low-power networks*

I. INTRODUCTION AND MOTIVATION

Contested electromagnetic environments fundamentally alter the design priorities of distributed communication systems. Modern field teams increasingly depend on edge computing nodes for sensing, local network interaction, and mission support [1]; however, these nodes operate under severe constraints imposed by RF detection, limited power availability, and physical exposure. Traditional high-power radios and centralized communication architectures provide bandwidth and

reliability, yet they increase electromagnetic signature and create single points of failure, particularly under deliberate jamming and spectrum denial conditions [2].

Existing mobile ad hoc network architectures and tactical communication systems prioritize throughput, latency, and routing efficiency, as reflected in prior wireless mesh network research, often assuming homogeneous node capability and persistent connectivity [3]. In contested environments, these assumptions break down. Survivability, detectability minimization, and energy persistence become dominant system-level objectives. This work reframes distributed mesh network design through a survivability-oriented lens. We propose a modular architecture that decouples computational payload nodes from ultra-low-power relay nodes and formalizes RF exposure as a quantifiable optimization objective within a graph-theoretic placement framework. By embedding stealth and energy efficiency directly into the design formulation, the system shifts from bandwidth maximization toward persistence under adversarial conditions.

II. SYSTEM REQUIREMENTS AND PROBLEM DEFINITION

Deployment in contested electromagnetic environments imposes both functional and non-functional requirements that directly constrain system architecture. Functionally, the network must support decentralized multi-hop communication without reliance on persistent centralized infrastructure, enable modular integration of sensing and computational payloads, and maintain connectivity under partial node failure or link degradation. Secure peer-to-peer control and encrypted data exchange are required to prevent compromise during operation. Non-functional requirements are dominated by survivability considerations, including minimizing electromagnetic detectability, bounding transmission power and duty cycles, extending battery endurance within size, weight, and power constraints, and resilience to spectrum denial or intermittent jamming. These requirements collectively frame the design problem as the development of a scalable, distributed edge network that balances capability with stealth and endurance.

Formally, the system must satisfy coverage and connectivity constraints while minimizing aggregate RF exposure and power consumption, motivating a constrained multi-objective optimization approach.

III. MODULAR EDGE NODE ARCHITECTURE

The proposed system employs a role-differentiated modular node architecture that decouples computational payload functions from low-power communication relays to reduce aggregate power consumption and electromagnetic exposure. Single-board computer nodes host sensing integration, local analytics, secure routing coordination, and cryptographic overlay services. Relay nodes are implemented on microcontroller-based platforms optimized for ultra-low-power operation and persistent availability. Their functionality is restricted to packet forwarding and mesh extension, enabling extended battery endurance and reduced detectability. By separating compute-intensive tasks from relaying, the architecture scales geographically without proportionally increasing power demand or exposure footprint. As illustrated in Figure 1, gateway and compute roles are physically and functionally separated.

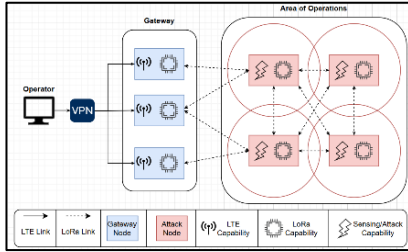


Figure 1. Role-differentiated gateway-compute node architecture in a contested electromagnetic environment.

The communication layer incorporates multiple communication transports, including cellular uplink on designated gateway nodes and long-range low-power mesh radios for distributed connectivity. This layered design supports survivability-driven requirements by enabling distributed mesh connectivity while allowing designated gateway nodes to provide higher-bandwidth uplink when infrastructure is available.

IV. MULTI-LAYER COMMUNICATION FRAMEWORK

The communication framework is designed as a hierarchical system that preserves connectivity across escalating levels of electromagnetic contention. Rather than optimizing for peak throughput, the architecture incorporates layered redundancy to allow for graceful degradation under adversarial conditions. When infrastructure is available, designated gateway nodes provide LTE-based uplink connectivity using Tailscale, providing relatively high bandwidth for standard remote operations. However, because LTE relies on centralized infrastructure that is easily detectable and vulnerable to jamming, it is treated as opportunistic rather than foundational.

In environments with limited infrastructure availability, connectivity is maintained through LoRa radios and Meshtastic, leveraging low-power wide-area networking principles designed for energy-efficient long-range communication [4]. This mesh layer prioritizes energy efficiency and low-profile

transmission, allowing for secure, SSH-like sessions even when cellular networks are denied. While this layer has lower bandwidth and higher latency, it provides the infrastructure-independent connectivity necessary for cyber sensing and remote tasking. Across these tiers, LTE and LoRa, the design explicitly manages the trade-offs between range, power consumption, and detectability. This layered approach is intended to preserve connectivity as electromagnetic contention escalates, shifting between communication modes without reconfiguring core application services.

V. NODE PLACEMENT MODELING AND TRADE-SPACE ANALYSIS

The effective deployment of a modular node network in contested environments requires formal mathematical treatment of coverage, connectivity, and stealth [5]. We frame this problem as a constrained optimization using integer programming to solve a specialized version of the set-cover problem [6]. The model's objective is to minimize the total number of devices required while simultaneously accounting for a stealth penalty. In this formulation, one variable serves as the decision variable determining whether a node is placed at a specific candidate location. To make sure the model remains realistic and computationally efficient for field operators, we utilize a binary sensing model, where an event is triggered simply by an object entering a node's detection space. While probabilistic models account for more real-world uncertainties, they increase complexity and computing time. By prioritizing the binary model, we meet the goal of creating a system that can be easily deployed into the field by cyber operators.

The model enforces several critical constraints to ensure mission success and device survivability. First, every identified demand location must be covered by at least one node, as we cannot be certain of the exact disposition or strength of enemy equipment. The model ensures a robust presence by covering all potential locations of enemy hardware. Second, the model defines an effective range for wireless cyber capabilities, ensuring that every device can reliably deliver effects without risking being out of range. Third, we implement a soft stealth constraint to ensure our devices are not easily detected from demand locations. By framing this as a "soft" constraint, the model allows closer placement only if a location is significantly more advantageous for the overall network. Finally, beyond simple coverage, we set the minimum edge connectivity to two [7]. This graph-theoretic requirement creates a communication network that is resilient to unpredictable circumstances, meaning it would take at least two devices to go out of service before any single device is disconnected from the others, as illustrated in Figure 2.

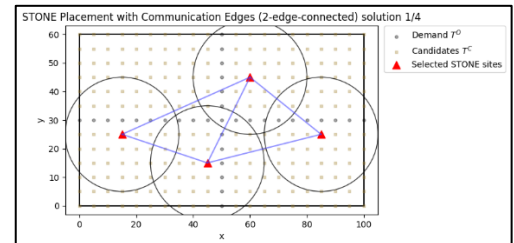


Figure 2. Representative 2-edge-connected node placement solution illustrating coverage regions and communication edges.

VI. PROTOTYPE DEVELOPMENT AND EARLY EXPERIMENTS

Our development followed a phased approach to evaluate whether open-source tools can provide greater anonymity, customization, and modularity at a significantly lower cost than traditional contractors. The Phase 1 MVP was constructed in a single day using Raspberry Pi single-board computers. These devices were equipped with LTE modem hats for internet backhaul and high-gain Wi-Fi adapters for wireless operations. To manage the network, we utilized a heterogeneous node structure where one specific device is equipped with a communication module. This allows a cyber operator to control the entire network via the internet from anywhere in the world, while the remaining nodes communicate via radio to stay hidden and reduce the risk of enemy detection.

Phase 2 validated a full demonstration, showcasing real operational capability from the platform. During this demo, the team successfully established remote access, performed network discovery, and interacted with representative IoT devices to verify command-and-control functionality across the deployed architecture. This demonstration confirmed that small-form-factor, open-source components could support secure remote interaction and distributed network operations within the proposed modular framework.

We are currently in Phase 3, focusing on mesh networking and electronic warfare survivability. By integrating LoRa radios and Meshtastic, we are enabling secure, SSH-like sessions over a low-power mesh network that can operate independently of LTE-based uplink connectivity. A major risk currently under investigation is power consumption, as transmission and reception can use up a majority of a node's energy. To mitigate this risk, we are working to limit unnecessary transmissions and better manage communication timing to conserve battery life. This phased progression ensures that the final product is not only concealable and battery-efficient but also fully fieldable in contested environments.

VII. OPERATIONAL IMPLICATIONS AND FUTURE WORK

The architectural and modeling decisions presented in this work have direct implications for distributed operations in contested or infrastructure-limited environments. By explicitly prioritizing survivability, bounded RF exposure, and energy persistence over throughput maximization, the proposed framework supports a shift toward resilient, distributed edge capabilities that degrade gracefully rather than fail catastrophically under electromagnetic contention. The asymmetric separation of compute and relay nodes enables flexible scaling of geographic coverage without proportionally increasing electromagnetic signature or power demand, while the layered communication hierarchy provides continuity across escalating levels of spectrum denial. Operationally, this structure supports decentralized sensing, local cyber assessment, and mission-relevant data exchange in scenarios where centralized command links are intermittent or unavailable. The modeling framework further enables planners to select deployment configurations aligned with mission-specific risk tolerance, balancing coverage density, network robustness, and exposure constraints within a formal trade-space rather than through ad hoc placement.

To extend the static optimization model into a mission-relevant planning tool, we incorporate a second analytical phase using discrete event simulation. While the integer program identifies feasible placement configurations under deterministic assumptions, the DES framework evaluates candidate solutions under operational uncertainty by estimating communication reliability, battery endurance, and the probability of maintaining cyber effects over extended mission durations. This layered evaluation allows planners to assess not only whether a network satisfies structural constraints, but whether it remains operationally viable over time. The simulation environment models stochastic transmission intervals, duty cycles, packet retransmissions, jamming-induced retries, and varying terrain-dependent link degradation. By propagating these effects through time, the framework estimates energy depletion rates and the likelihood of network partition under node failures or degraded links. This provides decision-makers with quantitative confidence bounds regarding expected mission persistence.

Future work will focus on empirical validation and model refinement across diverse terrain conditions, including open, wooded, and urban environments where attenuation, multipath effects, and structural obstruction influence connectivity and exposure. Field experiments will generate data to calibrate link reliability models, energy consumption profiles, and detectability proxies, enabling refinement of optimization weights and constraint bounds. Expansion of modular sensing and cyber payloads will extend the functional scope of edge nodes while preserving the decoupled relay structure. Parallel efforts will address power system improvements, including energy storage optimization and duty-cycle-aware management strategies to extend sustained operation. Beyond the immediate application domain, the survivability-oriented design paradigm developed here has broader relevance for cyber-physical systems operating under infrastructure uncertainty, disaster response networks, and distributed sensing architectures requiring persistent, low-profile operation.

Authors' Note: The views expressed herein are those of the authors and do not reflect the position of the United States Military Academy, the Department of the Army, or the Department of War.

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